

★ 14.4-6

Give an $O(n \lg n)$ -time algorithm to find the longest monotonically increasing subsequence of a sequence of n numbers. (*Hint*: The last element of a candidate subsequence of length i is at least as large as the last element of a candidate subsequence of length $i - 1$. Maintain candidate subsequences by linking them through the input sequence.)

14.5 Optimal binary search trees

Suppose that you are designing a program to translate text from English to Latvian. For each occurrence of each English word in the text, you need to look up its Latvian equivalent. You can perform these lookup operations by building a binary search tree with n English words as keys and their Latvian equivalents as satellite data. Because you will search the tree for each individual word in the text, you want the total time spent searching to be as low as possible. You can ensure an $O(\lg n)$ search time per occurrence by using a red-black tree or any other balanced binary search tree. Words appear with different frequencies, however, and a frequently used word such as *the* can end up appearing far from the root while a rarely used word such as *naumachia* appears near the root. Such an organization would slow down the translation, since the number of nodes visited when searching for a key in a binary search tree equals 1 plus the depth of the node containing the key. You want words that occur frequently in the text to be placed nearer the root.⁸ Moreover, some words in the text might have no Latvian translation,⁹ and such words would not appear in the binary search tree at all. How can you organize a binary search tree so as to minimize the number of nodes visited in all searches, given that you know how often each word occurs?

What you need is an *optimal binary search tree*. Formally, given a sequence $K = \langle k_1, k_2, \dots, k_n \rangle$ of n distinct keys such that $k_1 < k_2 < \dots < k_n$, build a binary search tree containing them. For each key k_i , you are given the probability p_i that any given search is for key k_i . Since some searches may be for values not in K , you also have $n + 1$ “dummy” keys

$d_0, d_1, d_2, \dots, d_n$ representing those values. In particular, d_0 represents all values less than k_1 , d_n represents all values greater than k_n , and for $i = 1, 2, \dots, n - 1$, the dummy key d_i represents all values between k_i and k_{i+1} . For each dummy key d_i , you have the probability q_i that a search corresponds to d_i . Figure 14.9 shows two binary search trees for a set of $n = 5$ keys. Each key k_i is an internal node, and each dummy key d_i is a leaf. Since every search is either successful (finding some key k_i) or unsuccessful (finding some dummy key d_i), we have

$$\sum_{i=1}^n p_i + \sum_{i=0}^n q_i = 1 . \quad (14.10)$$

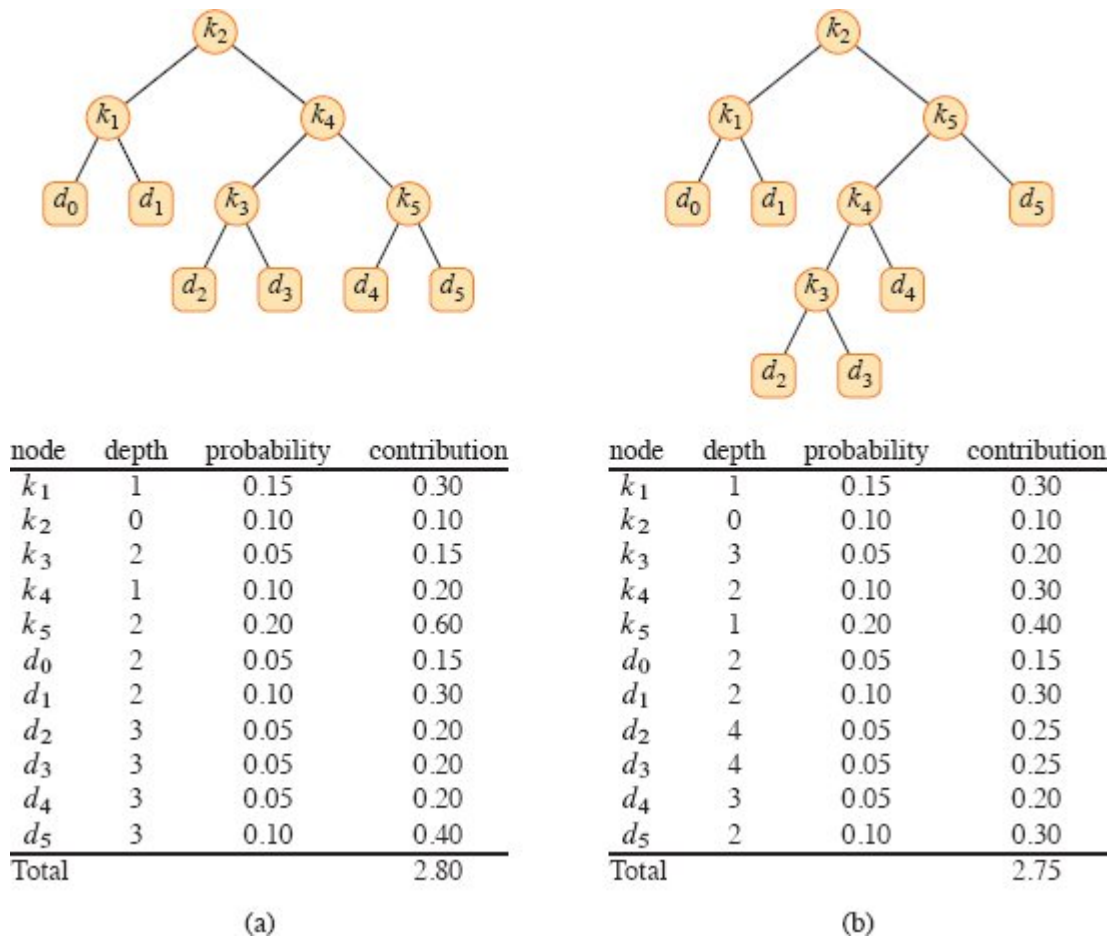


Figure 14.9 Two binary search trees for a set of $n = 5$ keys with the following probabilities:

i	0	1	2	3	4	5
p_i		0.15	0.10	0.05	0.10	0.20
q_i	0.05	0.10	0.05	0.05	0.05	0.10

(a) A binary search tree with expected search cost 2.80. **(b)** A binary search tree with expected search cost 2.75. This tree is optimal.

Knowing the probabilities of searches for each key and each dummy key allows us to determine the expected cost of a search in a given binary search tree T . Let us assume that the actual cost of a search equals the number of nodes examined, which is the depth of the node found by the search in T , plus 1. Then the expected cost of a search in T is

$$\begin{aligned}
E[\text{search cost in } T] &= \sum_{i=1}^n (\text{depth}_T(k_i) + 1) \cdot p_i + \sum_{i=0}^n (\text{depth}_T(d_i) + 1) \cdot q_i \\
&= 1 + \sum_{i=1}^n \text{depth}_T(k_i) \cdot p_i + \sum_{i=0}^n \text{depth}_T(d_i) \cdot q_i, \quad (14.11)
\end{aligned}$$

where depth_T denotes a node's depth in the tree T . The last equation follows from equation (14.10). Figure 14.9 shows how to calculate the expected search cost node by node.

For a given set of probabilities, your goal is to construct a binary search tree whose expected search cost is smallest. We call such a tree an *optimal binary search tree*. Figure 14.9(a) shows one binary search tree, with expected cost 2.80, for the probabilities given in the figure caption. Part (b) of the figure displays an optimal binary search tree, with expected cost 2.75. This example demonstrates that an optimal binary search tree is not necessarily a tree whose overall height is smallest. Nor does an optimal binary search tree always have the key with the greatest probability at the root. Here, key k_5 has the greatest search probability of any key, yet the root of the optimal binary search tree shown is k_2 . (The lowest expected cost of any binary search tree with k_5 at the root is 2.85.)

As with matrix-chain multiplication, exhaustive checking of all possibilities fails to yield an efficient algorithm. You can label the nodes of any n -node binary tree with the keys $k_1, k_2, \dots, k_n$ to construct a binary search tree, and then add in the dummy keys as leaves. In Problem 12-4 on page 329, we saw that the number of binary trees with n nodes is $\Omega(4^n/n^{3/2})$. Thus you would need to examine an exponential number of binary search trees to perform an exhaustive search. We'll see how to solve this problem more efficiently with dynamic programming.

Step 1: The structure of an optimal binary search tree

To characterize the optimal substructure of optimal binary search trees, we start with an observation about subtrees. Consider any subtree of a

binary search tree. It must contain keys in a contiguous range $k_i, \dots, k_j$, for some $1 \leq i \leq j \leq n$. In addition, a subtree that contains keys $k_i, \dots, k_j$ must also have as its leaves the dummy keys $d_{i-1}, \dots, d_j$.

Now we can state the optimal substructure: if an optimal binary search tree T has a subtree T' containing keys $k_i, \dots, k_j$, then this subtree T' must be optimal as well for the subproblem with keys $k_i, \dots, k_j$ and dummy keys $d_{i-1}, \dots, d_j$. The usual cut-and-paste argument applies. If there were a subtree T'' whose expected cost is lower than that of T' , then cutting T' out of T and pasting in T'' would result in a binary search tree of lower expected cost than T , thus contradicting the optimality of T .

With the optimal substructure in hand, here is how to construct an optimal solution to the problem from optimal solutions to subproblems. Given keys $k_i, \dots, k_j$, one of these keys, say k_r ($i \leq r \leq j$), is the root of an optimal subtree containing these keys. The left subtree of the root k_r contains the keys $k_i, \dots, k_{r-1}$ (and dummy keys $d_{i-1}, \dots, d_{r-1}$), and the right subtree contains the keys $k_{r+1}, \dots, k_j$ (and dummy keys $d_r, \dots, d_j$). As long as you examine all candidate roots k_r , where $i \leq r \leq j$, and you determine all optimal binary search trees containing $k_i, \dots, k_{r-1}$ and those containing $k_{r+1}, \dots, k_j$, you are guaranteed to find an optimal binary search tree.

There is one technical detail worth understanding about “empty” subtrees. Suppose that in a subtree with keys $k_i, \dots, k_j$, you select k_i as the root. By the above argument, k_i ’s left subtree contains the keys $k_i, \dots, k_{i-1}$: no keys at all. Bear in mind, however, that subtrees also contain dummy keys. We adopt the convention that a subtree containing keys $k_i, \dots, k_{i-1}$ has no actual keys but does contain the single dummy key d_{i-1} . Symmetrically, if you select k_j as the root, then k_j ’s right subtree contains the keys $k_{j+1}, \dots, k_j$. This right subtree contains no actual keys, but it does contain the dummy key d_j .

Step 2: A recursive solution

To define the value of an optimal solution recursively, the subproblem domain is finding an optimal binary search tree containing the keys $k_i, \dots, k_j$, where $i \geq 1, j \leq n$, and $j \geq i - 1$. (When $j = i - 1$, there is just the dummy key d_{i-1} , but no actual keys.) Let $e[i, j]$ denote the expected cost of searching an optimal binary search tree containing the keys $k_i, \dots, k_j$. Your goal is to compute $e[1, n]$, the expected cost of searching an optimal binary search tree for all the actual and dummy keys.

The easy case occurs when $j = i - 1$. Then the subproblem consists of just the dummy key d_{i-1} . The expected search cost is $e[i, i - 1] = q_{i-1}$.

When $j \geq i$, you need to select a root k_r from among $k_i, \dots, k_j$ and then make an optimal binary search tree with keys $k_i, \dots, k_{r-1}$ as its left subtree and an optimal binary search tree with keys $k_{r+1}, \dots, k_j$ as its right subtree. What happens to the expected search cost of a subtree when it becomes a subtree of a node? The depth of each node in the subtree increases by 1. By equation (14.11), the expected search cost of this subtree increases by the sum of all the probabilities in the subtree. For a subtree with keys $k_i, \dots, k_j$, denote this sum of probabilities as

$$w(i, j) = \sum_{l=i}^j p_l + \sum_{l=i-1}^j q_l. \quad (14.12)$$

Thus, if k_r is the root of an optimal subtree containing keys $k_i, \dots, k_j$, we have

$$e[i, j] = p_r + (e[i, r - 1] + w(i, r - 1)) + (e[r + 1, j] + w(r + 1, j)).$$

Noting that

$$w(i, j) = w(i, r - 1) + p_r + w(r + 1, j),$$

we rewrite $e[i, j]$ as

$$e[i, j] = e[i, r - 1] + e[r + 1, j] + w(i, j). \quad (14.13)$$

The recursive equation (14.13) assumes that you know which node k_r to use as the root. Of course, you choose the root that gives the lowest expected search cost, giving the final recursive formulation:

$$e[i, j] = \begin{cases} q_{i-1} & \text{if } j = i - 1, \\ \min \{e[i, r - 1] + e[r + 1, j] + w(i, j) : i \leq r \leq j\} & \text{if } i \leq j. \end{cases} \quad (14.14)$$

The $e[i, j]$ values give the expected search costs in optimal binary search trees. To help keep track of the structure of optimal binary search trees, define $root[i, j]$, for $1 \leq i \leq j \leq n$, to be the index r for which k_r is the root of an optimal binary search tree containing keys $k_i, \dots, k_j$. Although we'll see how to compute the values of $root[i, j]$, the construction of an optimal binary search tree from these values is left as Exercise 14.5-1.

Step 3: Computing the expected search cost of an optimal binary search tree

At this point, you may have noticed some similarities between our characterizations of optimal binary search trees and matrix-chain multiplication. For both problem domains, the subproblems consist of contiguous index subranges. A direct, recursive implementation of equation (14.14) would be just as inefficient as a direct, recursive matrix-chain multiplication algorithm. Instead, you can store the $e[i, j]$ values in a table $e[1 : n + 1, 0 : n]$. The first index needs to run to $n + 1$ rather than n because in order to have a subtree containing only the dummy key d_n , you need to compute and store $e[n + 1, n]$. The second index needs to start from 0 because in order to have a subtree containing only the dummy key d_0 , you need to compute and store $e[1, 0]$. Only the entries $e[i, j]$ for which $j \geq i - 1$ are filled in. The table $root[i, j]$ records the root of the subtree containing keys $k_i, \dots, k_j$ and uses only the entries for which $1 \leq i \leq j \leq n$.

One other table makes the dynamic-programming algorithm a little faster. Instead of computing the value of $w(i, j)$ from scratch every time you compute $e[i, j]$, which would take $\Theta(j - i)$ additions, store these

values in a table $w[1 : n + 1, 0 : n]$. For the base case, compute $w[i, i - 1] = q_{i-1}$ for $1 \leq i \leq n + 1$. For $j \geq i$, compute

$$w[i, j] = w[i, j - 1] + p_j + q_j . \quad (14.15)$$

Thus, you can compute the $\Theta(n^2)$ values of $w[i, j]$ in $\Theta(1)$ time each.

The OPTIMAL-BST procedure on the next page takes as inputs the probabilities $p_1, \dots, p_n$ and $q_0, \dots, q_n$ and the size n , and it returns the tables e and $root$. From the description above and the similarity to the MATRIX-CHAIN-ORDER procedure in Section 14.2, you should find the operation of this procedure to be fairly straightforward. The **for** loop of lines 2–4 initializes the values of $e[i, i - 1]$ and $w[i, i - 1]$. Then the **for** loop of lines 5–14 uses the recurrences (14.14) and (14.15) to compute $e[i, j]$ and $w[i, j]$ for all $1 \leq i \leq j \leq n$. In the first iteration, when $l = 1$, the loop computes $e[i, i]$ and $w[i, i]$ for $i = 1, 2, \dots, n$. The second iteration, with $l = 2$, computes $e[i, i + 1]$ and $w[i, i + 1]$ for $i = 1, 2, \dots, n - 1$, and so on. The innermost **for** loop, in lines 10–14, tries each candidate index r to determine which key k_r to use as the root of an optimal binary search tree containing keys $k_i, \dots, k_j$. This **for** loop saves the current value of the index r in $root[i, j]$ whenever it finds a better key to use as the root.

OPTIMAL-BST(p, q, n)

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1 let  $e[1 : n + 1, 0 : n]$ ,  $w[1 : n + 1, 0 : n]$ ,
   and  $root[1 : n, 1 : n]$  be new tables
2 for  $i = 1$  to  $n + 1$                                      // base cases
3    $e[i, i - 1] = q_{i-1}$                                    // equation (14.14)
4    $w[i, i - 1] = q_{i-1}$ 
5 for  $l = 1$  to  $n$ 
6   for  $i = 1$  to  $n - l + 1$ 
7      $j = i + l - 1$ 
8      $e[i, j] = \infty$ 
9      $w[i, j] = w[i, j - 1] + p_j + q_j$                  // equation (14.15)
10    for  $r = i$  to  $j$                                        // try all possible roots  $r$ 
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11       $t = e[i, r - 1] + e[r + 1, j] + w[i, j]$  // equation (14.14)
12      if  $t < e[i, j]$  // new minimum?
13           $e[i, j] = t$ 
14           $root[i, j] = r$ 
15 return  $e$  and  $root$ 

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Figure 14.10 shows the tables $e[i, j]$, $w[i, j]$, and $root[i, j]$ computed by the procedure OPTIMAL-BST on the key distribution shown in Figure 14.9. As in the matrix-chain multiplication example of Figure 14.5, the tables are rotated to make the diagonals run horizontally. OPTIMAL-BST computes the rows from bottom to top and from left to right within each row.

The OPTIMAL-BST procedure takes $\Theta(n^3)$ time, just like MATRIX-CHAIN-ORDER. Its running time is $O(n^3)$, since its **for** loops are nested three deep and each loop index takes on at most n values. The loop indices in OPTIMAL-BST do not have exactly the same bounds as those in MATRIX-CHAIN-ORDER, but they are within at most 1 in all directions. Thus, like MATRIX-CHAIN-ORDER, the OPTIMAL-BST procedure takes $\Omega(n^3)$ time.

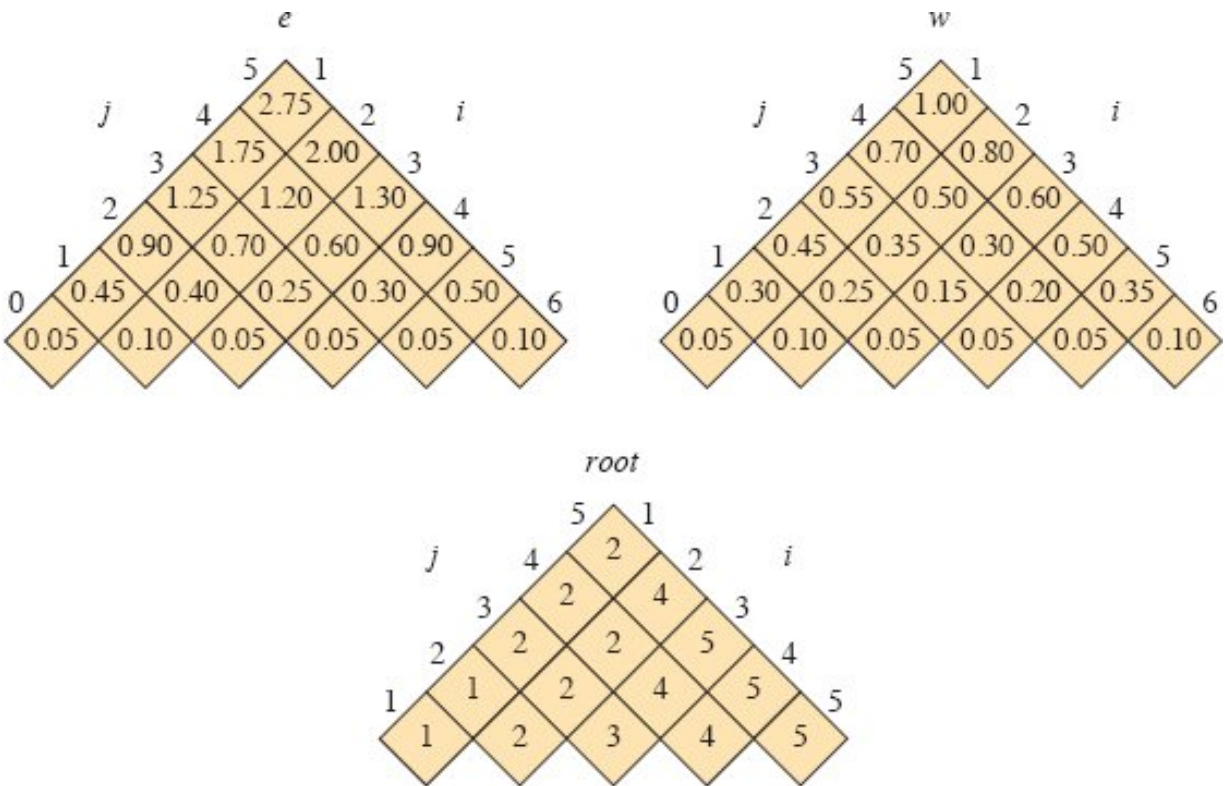


Figure 14.10 The tables $e[i, j]$, $w[i, j]$, and $root[i, j]$ computed by OPTIMAL-BST on the key distribution shown in Figure 14.9. The tables are rotated so that the diagonals run horizontally.

Exercises

14.5-1

Write pseudocode for the procedure **CONSTRUCT-OPTIMAL-BST**($root, n$) which, given the table $root[1 : n, 1 : n]$, outputs the structure of an optimal binary search tree. For the example in Figure 14.10, your procedure should print out the structure

- k_2 is the root
- k_1 is the left child of k_2
- d_0 is the left child of k_1
- d_1 is the right child of k_1
- k_5 is the right child of k_2

k_4 is the left child of k_5
 k_3 is the left child of k_4
 d_2 is the left child of k_3
 d_3 is the right child of k_3
 d_4 is the right child of k_4
 d_5 is the right child of k_5

corresponding to the optimal binary search tree shown in Figure 14.9(b).

14.5-2

Determine the cost and structure of an optimal binary search tree for a set of $n = 7$ keys with the following probabilities:

i	0	1	2	3	4	5	6	7
p_i		0.04	0.06	0.08	0.02	0.10	0.12	0.14
q_i	0.06	0.06	0.06	0.06	0.05	0.05	0.05	0.05

14.5-3

Suppose that instead of maintaining the table $w[i, j]$, you computed the value of $w(i, j)$ directly from equation (14.12) in line 9 of OPTIMAL-BST and used this computed value in line 11. How would this change affect the asymptotic running time of OPTIMAL-BST?

★ 14.5-4

Knuth [264] has shown that there are always roots of optimal subtrees such that $root[i, j - 1] \leq root[i, j] \leq root[i + 1, j]$ for all $1 \leq i < j \leq n$. Use this fact to modify the OPTIMAL-BST procedure to run in $\Theta(n^2)$ time.

Problems

14-1 Longest simple path in a directed acyclic graph

You are given a directed acyclic graph $G = (V, E)$ with real-valued edge weights and two distinguished vertices s and t . The *weight* of a path is the sum of the weights of the edges in the path. Describe a dynamic-